

IronBus on-disk format (little-endian, frozen v1): record frame and segment frame

Record frame (header 36 B, CRC32C over bytes [0..32))		
offset	field	type / width
0	magic = 0x4942	u16 (2 B)
2	version	u8 (1 B)
3	flags (compressed, has_key, ...)	u8 (1 B)
4	seq	u64 (8 B)
12	timestamp (ms)	u64 (8 B)
20	key_len	u32 (4 B)
24	hdr_len	u32 (4 B)
28	payload_len	u32 (4 B)
32	header_crc (CRC32C of [0..32))	u32 (4 B)
36	key bytes	key_len
...	headers	hdr_len
...	payload (optionally zstd, dict_id=0 in v1)	payload_len
...	body_crc (CRC32C of body; xxh3-64 too if payload > 64 KiB)	u32 (4 B)
...	total_len (whole record, enables reverse scan)	u32 (4 B)

Segment file (.log)	
Segment header (64 B, 4 KiB-aligned)	8-B magic "IRONBUS0", version, checksum_algo=0x1, segment_id u64, base_seq u64, base_offset u64, created_ts, header_crc
record 0	frame as above (seq = base_seq)
record 1 .. N-1	contiguous, record-aligned (NO 32 KiB block layer, NO fragments)
Segment footer (32 B)	record_count, last_seq, footer_crc (written on seal)

Derived sidecars (non-authoritative, rebuilt by recovery):
 .index sparse offset -> file position
 .tindex sparse timestamp -> offset
Naming: 20-digit zero-padded base offset. Limits: max record 16 MiB (to 1 GiB),
segment 64 MiB (8 MiB edge); a record never spans two segments.